

An agentic multi-agent system for realist evidence synthesis

Reproducing the analytic operations of a published realist review, with every claim traced to a verbatim quote.
Prepared for review; the system is under active development.

This report explains what the system is for, how it is built, and how well it reproduces a human realist review — Richmond et al.'s (2020) synthesis of educational interventions for clinical reasoning (28 papers). Verification is not a self-assigned grade: we transcribed Richmond's own published concepts, relations, and programme-theory chains into a reference key and checked, item by item, whether the system recovered them and from which quote. Every number below was taken from the run described here.

1 What the system is for

Realist synthesis asks not simply whether an intervention works, but *what* about it works, *for whom*, and *in what circumstances* (Pawson & Tilley, 1997). Its unit of analysis is the **Context–Mechanism–Outcome configuration** (CMOC): within a context, an intervention offers a resource, a learner responds, and an outcome follows. Assembling these across a body of literature and drawing them into a programme theory is slow expert work.

The goal is a system that performs these operations with a human in charge at every judgement point, and that is not tied to one review method: a method-agnostic core with realist synthesis added as a plug-in. Success is defined narrowly — can the system recover the contexts, mechanisms, outcomes, and programme theory a careful human team published, with each element auditable against the source.

2 Entities and relationships: definitions and types

The knowledge graph is built from two kinds of object. An **entity** is a single realist element extracted from a paper and grounded in a verbatim quote; every entity has exactly one of five types. A **relationship** is a directed, typed causal link between two entities; only five predicates are allowed, each with a fixed domain and range, so the graph stays a coherent $C \rightarrow M \rightarrow O$ structure rather than free text.

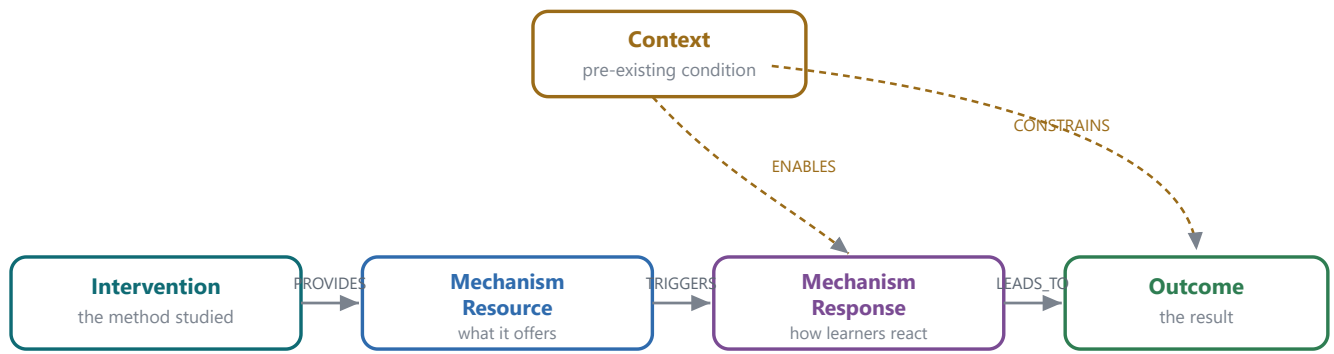


Figure 1. The realist ontology: five entity types and the five directed predicates (with their fixed domain→range) that connect them into a C→M→O configuration.

Table 1. The five relationship predicates and where each may apply.

PREDICATE	FROM (DOMAIN)	TO (RANGE)
PROVIDES	Intervention	Mechanism Resource / Response
TRIGGERS	Intervention / Mechanism Resource	Mechanism Response / Outcome
ENABLES	Context	Mechanism Response / Outcome
LEADS_TO	Mechanism Response	Mechanism Response / Outcome
CONSTRAINS	Context	Outcome

A link whose endpoints break these rules is not silently kept — it is demoted to an untyped candidate and surfaced for human review. On this run the graph holds **180 concept-level entities** and **439 typed relations**.

3 Architecture and algorithm

The system runs as a staged pipeline of specialised agents over one shared database — the single source of truth. Narrow agents give more faithful output than one agent asked to do everything, and they mirror the division of labour in a review team. A human sign-off sits at each stage that needs judgement.

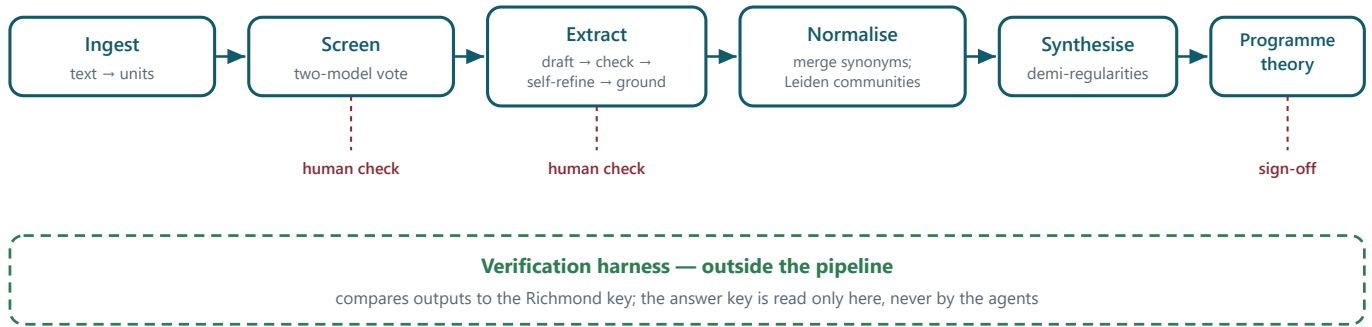


Figure 2. The pipeline. Each stage is an agent writing to the shared database; humans sign off at the marked points. The benchmark comparison sits outside the system so the agents never see the answer key.

Two design choices are worth naming. First, extraction is **self-correcting**: the extractor drafts, an independent checker in a *different* model family critiques the draft, and the extractor revises before anything is stored — echoing Richmond's practice of a second reviewer checking every paper. Self-refinement of this kind is established to improve model output (Madaan et al., 2023; Shinn et al., 2023). Second, higher-level concepts are **discovered, not pre-listed**: entities are clustered by Leiden community detection (Traag et al., 2019), the method Microsoft's GraphRAG uses (Edge et al., 2024), and each community is summarised into one "big concept". One empirical caution shaped the design — injecting the full programme theory into the per-paper extraction prompt was found to **reduce** how many configurations were recovered, so that guidance is kept light and steers naming and synthesis, not the first read.

4 Does the system follow Richmond's workflow?

We mapped each step of Richmond's human process onto an agent. The table records whether the step was reproduced and what it produced on this run.

Table 2. Richmond's human workflow beside the system, with the result at each step.

RICHMOND STEP (HUMAN)	SYSTEM AGENT	FOLLOWED	RESULT ON THIS RUN
Write the initial programme theory	Protocol & initial-theory agent	Yes	Public-theory seed; steers synthesis, not the per-paper read
Search databases	—	Partial	Out of scope for the benchmark; the 28-paper corpus is fixed to Richmond's
Screen abstracts, then full texts	Two-stage screening agent + human	Yes	Reproduced the inclusion set — 100% sensitivity after human review
Two reviewers extract CMOCs	Extraction agent (self-refine)	Yes	91 configurations; 96.9% of quotes resolve to the source text
Second reviewer checks each paper	Independent checker (different model)	Yes	Every study re-checked; disagreements surfaced, not overwritten
Synthesise across studies	Normalisation + Leiden communities	Yes	180 concepts, 8 big concepts, 36 demi-regularities
Build programme theory by context	Theory synthesis & refinement	Yes	Seven context sections (see §6)
Report with an audit trail	Reporting & audit agent	Yes	Outputs plus full provenance for RAMESES reporting (Wong et al., 2013)

5 How verification works — and what it is *not*

Richmond's paper is prose, not a scored test. So the standard is built **from that prose**: we read their Results §3.2 and Figures 2 and 3 and transcribed, by hand, the analytic content they published — **47 concepts** (their contexts, interventions, mechanisms and outcomes), **40 causal relations** among them, and their **5 context-specific programme-theory chains** — each tagged with its location in the paper. This transcription is the reference key.

Verification then compares the system's output to that key **one item at a time**: for each concept Richmond describes, is there an extracted concept expressing the same thing, and if so from which study and quote? A number such as "43 of 47" is therefore not a grade the system gave itself — it is a count of how many of Richmond's published items were recovered, and **every count is traceable** to a page in Richmond and a verbatim quote in a source paper (§6). Where the semantic match is a

judgement call, it is made by a separate model and flagged for a human to ratify; it is never treated as settled.

6 Results

The centre of the result is the content match below, which can be checked directly against Richmond's paper. The summary figures follow it.

6.1 Content recovered, with evidence

A representative selection across the five entity types (the full 47-item list is in `Richmond_Correspondence.md`). "Different role" marks a concept the system recovered but filed under a different realist role than Richmond — surfaced honestly, not hidden.

Table 3. Richmond's published elements beside what the system recovered, with the source quote.

RICHMOND ELEMENT · LOCATION	RECOVERED AS · STUDY · QUOTE
Context low clinical knowledge / cannot yet use knowledge Fig 2, pp.712–14	novices not using non-analytic reasoning S006, S013, S016, S018 <i>"they are not yet likely to have well-developed illness scripts."</i>
Context high clinical knowledge §3.2	clinically different expertise levels S001, S009, S022 ▶ different role <i>"Scores were compared between groups of different levels of expertise: surgeons, residents, interns..."</i>
Intervention expert's reasoning explicitly revealed §3.2.2, p.714	case-based clinical reasoning seminar S001, S003, S004, S005 <i>"...designed to bring students insight into cognitive features of their reasoning..."</i>
Intervention real-life scenarios incl. simulation §3.2.3, p.714	diagnostic decision-tree virtual-patient cases S011, S015 <i>"...software can be used to create VP cases that give virtual experience of these real-time clinical reasoning techniques."</i>
Resource multiple relevant resources Fig 2, p.713	multiple tasks and acutely unwell patients S011, S015, S022, S023 ▶ different role <i>"...the students were faced with multiple tasks and acutely unwell patients, and therefore had to prioritise."</i>
Response cognitive load is increased §3.2.4, p.716	cognitive-load overwhelm S016, S024, S025, S028 <i>"...may simply overwhelm an individual's cognitive load without any effect on outcomes."</i>
Response stress §3.2.4, p.716	stress from competing patient and pager demands S015 <i>"It was stressful. When you have a sick patient and then the page goes off, that was stressful."</i>
Response frustrated §3.2.2, p.714	student frustration S022 <i>"8 of 17 found the game frustrating (compared to 1 of 11 trainees and 1 of 6 experts)."</i>
Outcome high diagnostic accuracy §3.2.2, p.714	improved diagnostic accuracy S003, S004, S005, S019 <i>"Mean diagnostic accuracy scores were significantly higher in the analytic reasoning group than in the control group."</i>
Outcome more complete illness scripts §3.2.3, p.714	illness-script development for diagnosis S023 <i>"pools of similar cases are required so that students can properly develop their illness scripts..."</i>

6.2 Summary of the match

100%

Screening sensitivity (after human review)

43 / 47

Richmond concepts recovered (37 same role, 6 different role)

39 / 40

Causal relations recovered at the pattern level

96.9%

Extracted elements grounded in an exact source quote

Relations at the strictest level (both endpoints the exact same concept) are **0 of 40**; with one exact endpoint, **22 of 40**. We report all three levels rather than the flattering one: for a realist review the causal *pattern* (e.g. *Context constrains Outcome*) is the appropriate target, and it is nearly complete.

6.3 Programme-theory chains

Richmond's five context-specific chains, and how closely the system's programme theory corresponds to each (judged by a separate model; a human expert rating is still required before any publication claim).

Table 4. Correspondence of the system's programme theory to Richmond's five chains.

CHAIN · RICHMOND CONTEXT	CORRESPONDENCE
PTS1 low domain knowledge	0.89 — strong
PTS4 negative coping / low self-efficacy	0.59 — moderate
PTS2 high domain knowledge	0.50 — partial
PTS5 mixed-knowledge group	0.39 — limited
PTS3 positive coping / self-efficacy	0.32 — limited

7 Honest assessment

What works. The system reproduces the inclusion decisions, recovers most of the content Richmond published, and grounds nearly every element in a verbatim quote — so the output can be audited rather than trusted on faith. The causal patterns are almost all present, and the big concepts the graph finds on its own (guided reasoning practice, self-explanation, testing-enhanced study, and so on) are recognisably the mechanisms a human would name. Its strongest programme-theory chain, for low-knowledge novices, corresponds closely to Richmond's.

What is not there yet. Four of Richmond's 47 elements were missed on this run, including the mixed-knowledge-group context and two affective responses. Six matches sit under a different realist

role than Richmond assigned, and a few of those are generous — they are flagged for human ratification, not counted as clean. Exact-concept relation matching is low, and the programme-theory correspondence is model-judged and awaits an expert rating. These are specific, understood gaps, not diffuse weakness.

Is the current result strong enough to keep going? We think yes. On a genuinely hard expert task the system already recovers most of a published human review with traceable evidence, and where it falls short the reasons are concrete. Three mechanisms requested for this project — seed-guided extraction, graph-aware extraction, and a human-feedback-to-few-shot loop — are now built in and were active on this run; refining them is the next step.

8 References

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All figures and counts are generated from the run described here (26 of 28 papers extracted; two are abstract-only and correctly yielded nothing). The interactive knowledge graph is available in the review console. This is a progress report on a system under active development.